

Grouping Hosts into Subnets

Task 1

Which IP addresses can be given to hosts in the following subnets?

Which IP addresses can be given to hosts in the following subnets?

Start	192.168.17.32 / 27	192.168.1.24 / 29	192.168.1.96 / 27	192.168.1.64 / 28
	192.168.17.31	192.168.1.23	192.168.1.97	192.168.1.79
	192.168.17.35	192.168.1.28	192.168.1.32	192.168.1.60
	192.168.17.63	192.168.1.31	192.168.1.64	192.168.1.128
	192.168.17.72	192.168.1.32	192.168.1.98	192.168.1.55
	192.168.17.8	192.168.1.29	192.168.1.91	192.168.1.80
	192.168.17.55	192.168.1.30	192.168.1.127	192.168.1.90
	192.168.17.64	192.168.1.24	192.168.1.126	192.168.1.77
	192.168.17.72	192.168.1.26	192.168.1.224	192.168.1.68
	192.168.17.8	192.168.1.25	192.168.1.223	192.168.1.72
	192.168.17.15	192.168.1.29	192.168.1.96	192.168.1.66
	192.168.17.44	192.168.1.16	192.168.1.100	192.168.1.65

Which IP addresses can be given to hosts in the following subnets?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

Start	192.168.17.32 / 27	192.168.1.24 / 29	192.168.1.96 / 27	192.168.1.64 / 28
	192.168.17.31	192.168.1.23	192.168.1.97	192.168.1.79
	192.168.17.35	192.168.1.28	192.168.1.32	192.168.1.60
	192.168.17.63	192.168.1.31	192.168.1.64	192.168.1.128
	192.168.17.72	192.168.1.32	192.168.1.98	192.168.1.55
	192.168.17.8	192.168.1.29	192.168.1.91	192.168.1.80
	192.168.17.55	192.168.1.30	192.168.1.127	192.168.1.90
	192.168.17.64	192.168.1.24	192.168.1.126	192.168.1.77
	192.168.17.72	192.168.1.26	192.168.1.224	192.168.1.68
	192.168.17.8	192.168.1.25	192.168.1.223	192.168.1.72
	192.168.17.15	192.168.1.29	192.168.1.96	192.168.1.66
	192.168.17.44	192.168.1.16	192.168.1.100	192.168.1.65

Which IP addresses can be given to hosts in the following subnets?

128 64 32 16 8 4 2 1
7 6 5 4 3 2 1 0

Start	192.168.17.32 / 27	192.168.1.24 / 29	192.168.1.96 / 27	192.168.1.64 / 28
Delta	32			
Subnet	192.168.17.32			
1 st	192.168.17.33			
Last	192.168.17.62			
B'cast	192.168.17.63			
Next	192.168.17.64			
	192.168.17.31	192.168.1.23	192.168.1.97	192.168.1.79
	192.168.17.35	192.168.1.28	192.168.1.32	192.168.1.60
	192.168.17.63	192.168.1.31	192.168.1.64	192.168.1.128
	192.168.17.71	192.168.1.32	192.168.1.98	192.168.1.55
	192.168.17.8	192.168.1.29	192.168.1.91	192.168.1.80
	192.168.17.55	192.168.1.30	192.168.1.127	192.168.1.90
	192.168.17.64	192.168.1.24	192.168.1.126	192.168.1.77
	192.168.17.72	192.168.1.26	192.168.1.224	192.168.1.68
	192.168.17.8	192.168.1.25	192.168.1.223	192.168.1.72
	192.168.17.15	192.168.1.29	192.168.1.96	192.168.1.66
	192.168.17.44	192.168.1.16	192.168.1.100	192.168.1.65

Which IP addresses can be given to hosts in the following subnets?

128 64 32 16 8 4 2 1
7 6 5 4 3 2 1 0

Start	192.168.17.32 / 27	192.168.1.24 / 29	192.168.1.96 / 27	192.168.1.64 / 28
Delta	32	8		
Subnet 1 st	192.168.17.32	192.168.1.24		
Last	192.168.17.62	192.168.1.30		
B'cast	192.168.17.63	192.168.1.31		
Next	192.168.17.64	192.168.1.32		
	192.168.17.31	192.168.1.23	192.168.1.97	192.168.1.79
	192.168.17.35	192.168.1.28	192.168.1.32	192.168.1.60
	192.168.17.63	192.168.1.31	192.168.1.64	192.168.1.128
	192.168.17.72	192.168.1.32	192.168.1.98	192.168.1.55
	192.168.17.8	192.168.1.23	192.168.1.91	192.168.1.80
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	192.168.17.72	192.168.1.26	192.168.1.224	192.168.1.68
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	192.168.17.44	192.168.1.16	192.168.1.100	192.168.1.65

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Start	192.168.17.32 / 27	192.168.1.24 / 29	192.168.1.96 / 27	192.168.1.64 / 28
Delta	32	8	32	
Subnet 1 st	192.168.17.32	192.168.1.24	192.168.1.96	
Last	192.168.17.62	192.168.1.30	192.168.1.126	
B'cast	192.168.17.63	192.168.1.31	192.168.1.127	
Next	192.168.17.64	192.168.1.32	192.168.1.128	
	192.168.17.31	192.168.1.23	192.168.1.97	192.168.1.79
	192.168.17.35	192.168.1.28	192.168.1.32	192.168.1.60
	192.168.17.63	192.168.1.31	192.168.1.64	192.168.1.128
	192.168.17.72	192.168.1.32	192.168.1.98	192.168.1.55
	192.168.17.8	192.168.1.29	192.168.1.91	192.168.1.80
	192.168.17.55	192.168.1.30	192.168.1.127	192.168.1.90
	192.168.17.64	192.168.1.24	192.168.1.126	192.168.1.77
	192.168.17.72	192.168.1.26	192.168.1.224	192.168.1.68
	192.168.17.8	192.168.1.25	192.168.1.223	192.168.1.72
	192.168.17.15	192.168.1.29	192.168.1.96	192.168.1.66
	192.168.17.44	192.168.1.16	192.168.1.100	192.168.1.65

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7	6	5	4	3	2	1	0

Start	192.168.17.32 / 27	192.168.1.24 / 29	192.168.1.96 / 27	192.168.1.64 / 28
Delta	32	8	32	16
Subnet	192.168.17.32	192.168.1.24	192.168.1.96	192.168.1.64
1 st	192.168.17.33	192.168.1.25	192.168.1.97	192.168.1.65
Last	192.168.17.62	192.168.1.30	192.168.1.126	192.168.1.78
B'cast	192.168.17.63	192.168.1.31	192.168.1.127	192.168.1.79
Next	192.168.17.64	192.168.1.32	192.168.1.128	192.168.1.80
	192.168.17.31	192.168.1.23	192.168.1.97	192.168.1.79
	192.168.17.35	192.168.1.28	192.168.1.32	192.168.1.60
	192.168.17.63	192.168.1.31	192.168.1.64	192.168.1.128
	192.168.17.72	192.168.1.32	192.168.1.98	192.168.1.55
	192.168.17.8	192.168.1.29	192.168.1.91	192.168.1.80
	192.168.17.55	192.168.1.30	192.168.1.127	192.168.1.90
	192.168.17.64	192.168.1.24	192.168.1.126	192.168.1.77
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	192.168.17.15	192.168.1.29	192.168.1.96	192.168.1.66
	192.168.17.44	192.168.1.16	192.168.1.100	192.168.1.65

Task 2

Which IP addresses can be given to hosts in the following subnets?

Which IP addresses can be given to hosts in the following subnets?

128 64 32 16 8 4 2 1
7 6 5 4 3 2 1 0

Start	172.16.48.0 /20	172.16.20.0 /23	172.16.32.0 /22	172.16.32.0 /21	172.16.32.0 /19
	172.17.0.0	172.16.0.1	172.16.0.1	172.17.0.0	172.16.45.154
	172.31.255.254	172.16.20.1	172.16.35.3	172.31.255.254	172.16.55.254
	172.16.48.25	172.16.21.0	172.16.33.255	172.16.39.0	172.16.55.255
	172.16.52.255	172.16.21.33	172.16.34.0	172.16.33.255	172.16.48.100
	172.16.63.0	172.16.20.255	172.16.32.0	172.16.30.255	172.16.30.255
	172.16.49.0	172.16.20.254	172.16.35.67	172.16.32.187	172.16.61.183
	172.16.48.2	172.16.20.74	172.16.32.192	172.16.36.88	172.16.39.27
	172.16.50.255	172.16.31.1	172.16.34.224	172.16.37.202	172.16.37.256

Which IP addresses can be given to hosts in the following subnets?

128 64 32 16 8 4 2 1
7 6 5 4 3 2 1 0

Start	172.16.48.0 /20	172.16.20.0 /23	172.16.32.0 /22	172.16.32.0 /21	172.16.32.0 /19
Delta	16	2	4	8	32
Octet	3 rd	3 rd	3 rd	3 rd	3 rd
Subnet	172.16.48.0	172.16.20.0	172.16.32.0	172.16.32.0	172.16.32.0
1 st	172.16.48.1	172.16.20.1	172.16.32.1	172.16.32.1	172.16.31.1
Last	172.16.63.254	172.16.21.254	172.16.35.254	172.16.39.254	172.16.63.254
B'cast	172.16.63.255	172.16.21.255	172.16.35.255	172.16.39.255	172.16.63.255
Next	172.16.64.0	172.16.22.0	172.16.36.0	172.16.40.0	172.16.64.0
	172.17.0.0	172.16.0.1	172.16.0.1	172.17.0.0	172.16.45.154
	172.31.255.254	172.16.20.1	172.16.35.3	172.31.255.254	172.16.55.254
	172.16.48.25	172.16.21.0	172.16.33.255	172.16.39.0	172.16.55.255
	172.16.52.255	172.16.21.33	172.16.34.0	172.16.33.255	172.16.48.100
	172.16.63.0	172.16.20.255	172.16.32.0	172.16.30.255	172.16.30.255
	172.16.49.0	172.16.20.254	172.16.35.67	172.16.32.187	172.16.61.183
	172.16.48.2	172.16.20.74	172.16.32.192	172.16.36.88	172.16.39.27
	172.16.50.255	172.16.31.1	172.16.34.224	172.16.37.202	172.16.37.256

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Start	172.16.48.0 /20	172.16.20.0 /23	172.16.32.0 /22	172.16.32.0 /21	172.16.32.0 /19
Delta	16	2	4	8	32
Octet	3 rd	3 rd	3 rd	3 rd	3 rd
Subnet	172.16.48.0	172.16.20.0	172.16.32.0	172.16.32.0	172.16.32.0
1 st	172.16.48.1	172.16.20.1	172.16.32.1	172.16.32.1	172.16.31.1
Last	172.16.63.254	172.16.21.254	172.16.35.254	172.16.39.254	172.16.63.254
B'cast	172.16.63.255	172.16.21.255	172.16.35.255	172.16.39.255	172.16.63.255
Next	172.16.64.0	172.16.22.0	172.16.36.0	172.16.40.0	172.16.64.0

172.17.0.0	172.16.0.1	172.16.0.1	172.17.0.0	172.16.45.154
172.31.255.254	172.16.20.1	172.16.35.3	172.31.255.254	172.16.55.254
172.16.48.25	172.16.21.0	172.16.33.255	172.16.39.0	172.16.55.255
172.16.52.255	172.16.21.33	172.16.34.0	172.16.33.255	172.16.48.100
172.16.63.0	172.16.20.255	172.16.32.0	172.16.30.255	172.16.30.255
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172.16.48.2	172.16.20.74	172.16.32.192	172.16.36.88	172.16.39.27
172.16.50.255	172.16.31.1	172.16.34.224	172.16.37.202	172.16.37.256

Which IP addresses can be given to hosts in the following subnets?

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Start	172.16.48.0 /20	172.16.20.0 /23	172.16.32.0 /22	172.16.32.0 /21	172.16.32.0 /19
Delta	16	2	4	8	32
Octet	3 rd	3 rd	3 rd	3 rd	3 rd
Subnet	172.16.48.0	172.16.20.0	172.16.32.0	172.16.32.0	172.16.32.0
1 st	172.16.48.1	172.16.20.1	172.16.32.1	172.16.32.1	172.16.31.1
Last	172.16.63.254	172.16.21.254	172.16.35.254	172.16.39.254	172.16.63.254
B'cast	172.16.63.255	172.16.21.255	172.16.35.255	172.16.39.255	172.16.63.255
Next	172.16.64.0	172.16.22.0	172.16.36.0	172.16.40.0	172.16.64.0

172.17.0.0	172.16.0.1	172.16.0.1	172.17.0.0	172.16.45.154
172.31.255.254	172.16.20.1	172.16.35.3	172.31.255.254	172.16.55.254
172.16.48.25	172.16.21.0	172.16.33.255	172.16.39.0	172.16.55.255
172.16.52.255	172.16.21.33	172.16.34.0	172.16.33.255	172.16.48.100
172.16.63.0	172.16.20.255	172.16.32.0	172.16.30.255	172.16.30.255
172.16.49.0	172.16.20.254	172.16.35.67	172.16.32.187	172.16.61.183
172.16.48.2	172.16.20.74	172.16.32.192	172.16.36.88	172.16.39.27
172.16.50.255	172.16.31.1	172.16.36.224	172.16.37.202	172.16.37.256

Which IP addresses can be given to hosts in the following subnets?

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7 6 5 4 3 2 1 0

Start	172.16.48.0 /20	172.16.20.0 /23	172.16.32.0 /22	172.16.32.0 /21	172.16.32.0 /19
Delta	16	2	4	8	32
Octet	3 rd	3 rd	3 rd	3 rd	3 rd
Subnet	172.16.48.0	172.16.20.0	172.16.32.0	172.16.32.0	172.16.32.0
1 st	172.16.48.1	172.16.20.1	172.16.32.1	172.16.32.1	172.16.31.1
Last	172.16.63.254	172.16.21.254	172.16.35.254	172.16.39.254	172.16.63.254
B'cast	172.16.63.255	172.16.21.255	172.16.35.255	172.16.39.255	172.16.63.255
Next	172.16.64.0	172.16.22.0	172.16.36.0	172.16.40.0	172.16.64.0

172.17.0.0	172.16.0.1	172.16.0.1	172.17.0.0	172.16.45.154
172.31.255.254	172.16.20.1	172.16.35.3	172.31.255.254	172.16.55.254
172.16.48.25	172.16.21.0	172.16.33.255	172.16.39.0	172.16.55.255
172.16.52.255	172.16.21.33	172.16.34.0	172.16.33.255	172.16.48.100
172.16.63.0	172.16.20.255	172.16.32.0	172.16.30.255	172.16.30.255
172.16.49.0	172.16.20.254	172.16.35.67	172.16.32.187	172.16.61.183
172.16.48.2	172.16.20.74	172.16.32.192	172.16.36.88	172.16.39.27
172.16.50.255	172.16.31.1	172.16.34.224	172.16.37.202	172.16.37.256

Which IP addresses belong to the following subnets?

128 64 32 16 8 4 2 1
7 6 5 4 3 2 1 0

Start	172.16.48.0 /20	172.16.20.0 /23	172.16.32.0 /22	172.16.32.0 /21	172.16.32.0 /19
Delta	16	2	4	8	32
Octet	3 rd	3 rd	3 rd	3 rd	3 rd
Subnet	172.16.48.0	172.16.20.0	172.16.32.0	172.16.32.0	172.16.32.0
1 st	172.16.48.1	172.16.20.1	172.16.32.1	172.16.32.1	172.16.31.1
Last	172.16.63.254	172.16.21.254	172.16.35.254	172.16.39.254	172.16.63.254
B'cast	172.16.63.255	172.16.21.255	172.16.35.255	172.16.39.255	172.16.63.255
Next	172.16.64.0	172.16.22.0	172.16.36.0	172.16.40.0	172.16.64.0

172.17.0.0	172.16.0.1	172.16.0.1	172.17.0.0	172.16.45.154
172.31.255.254	172.16.20.1	172.16.35.3	172.31.255.254	172.16.55.254
172.16.48.25	172.16.21.0	172.16.33.255	172.16.39.0	172.16.55.255
172.16.52.255	172.16.21.33	172.16.34.0	172.16.33.255	172.16.64.100
172.16.63.0	172.16.20.255	172.16.32.0	172.16.30.255	172.16.30.255
172.16.49.0	172.16.20.254	172.16.35.67	172.16.32.187	172.16.61.183
172.16.48.2	172.16.20.74	172.16.32.192	172.16.36.88	172.16.31.27
172.16.50.255	172.16.31.1	172.16.34.224	172.16.37.202	172.16.37.256

Task 3

- Based on the subnet mask, which IP addresses could be grouped together into the same subnet?
- What would their subnet ID be?
- Is the address a legitimate host IP, a broadcast, or the subnet ID?

Based on the subnet mask, which IP addresses could be grouped together into the same subnet?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or the subnet ID?

/24	/28	/25	/29
192.168.2.0	192.168.1.0	10.2.2.128	172.16.10.0
192.168.5.1	192.168.1.1	10.2.2.128	172.16.10.24
192.168.2.255	192.168.1.13	10.2.2.190	172.16.10.31
192.168.2.28	192.168.1.14	10.2.2.192	172.16.10.30
192.168.3.166	192.168.1.15	10.2.2.254	172.16.10.8
192.168.2.20	192.168.1.16	10.2.2.255	172.16.10.26
192.168.4.254	192.168.1.17	10.2.2.256	172.16.10.25

Based on the subnet mask, which IP addresses could be grouped together into the same subnet?
What would their subnet ID be?
Is the address a legitimate host IP, a broadcast, or the subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

/24	/28	/25	/29
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Delta	1
Octet	3 rd
Example	0.0 – 0.255
Ranges	1.0 – 1.255
	2.0 – 2.255
	3.0 – 3.255

Subnet – even
1st – odd
Last – even
B’cast - odd

192.168.2.0	192.168.1.0	10.2.2.128	172.16.10.0
192.168.5.1	192.168.1.1	10.2.2.128	172.16.10.24
192.168.2.255	192.168.1.13	10.2.2.190	172.16.10.31
192.168.2.28	192.168.1.14	10.2.2.192	172.16.10.30
192.168.3.166	192.168.1.15	10.2.2.254	172.16.10.8
192.168.2.20	192.168.1.16	10.2.2.255	172.16.10.26
192.168.4.254	192.168.1.17	10.2.2.256	172.16.10.25

Based on the subnet mask, which IP addresses could be grouped together into the same subnet?
What would their subnet ID be?
Is the address a legitimate host IP, a broadcast, or the subnet ID?

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7	6	5	4	3	2	1	0

	/24	/28	/25	/29
Delta	1	16		
Octet	3 rd	4 th		
Example	0 – 255	0 – 15		
Ranges	0.0 – 0.255	16 – 31		
	1.0 – 1.255	32 – 47		
	2.0 – 2.255	128 – 143		
	3.0 – 3.255	160 – 175		
		...		
Subnet – even	192.168.2.0	192.168.1.0	10.2.2.128	172.16.10.0
1 st – odd	192.168.5.1	192.168.1.1	10.2.2.128	172.16.10.24
Last – even	192.168.2.255	192.168.1.13	10.2.2.190	172.16.10.31
B'cast - odd	192.168.2.28	192.168.1.14	10.2.2.192	172.16.10.30
	192.168.3.166	192.168.1.15	10.2.2.254	172.16.10.8
	192.168.2.20	192.168.1.16	10.2.2.255	172.16.10.26
	192.168.4.254	192.168.1.17	10.2.2.256	172.16.10.25

Based on the subnet mask, which IP addresses could be grouped together into the same subnet?
What would their subnet ID be?
Is the address a legitimate host IP, a broadcast, or the subnet ID?

128 64 32 16 8 4 2 1
7 6 5 4 3 2 1 0

/24 /28 /25 /29

Delta	1	16
Octet	3 rd	4 th
Example	0 – 255	0 – 15
Ranges	0.0 – 0.255	16 – 31
	1.0 – 1.255	32 – 47
	2.0 – 2.255	128 - 143
	3.0 – 3.255	160 - 175
		...

Subnet – even
1st – odd
Last – even
B’cast - odd

192.168.2.0	192.168.1.0	10.2.2.128	172.16.10.0
192.168.5.1	192.168.1.1	10.2.2.128	172.16.10.24
192.168.2.255	192.168.1.13	10.2.2.190	172.16.10.31
192.168.2.28	192.168.1.14	10.2.2.192	172.16.10.30
192.168.3.166	192.168.1.15	10.2.2.254	172.16.10.8
192.168.2.20	192.168.1.16	10.2.2.255	172.16.10.26
192.168.4.254	192.168.1.17	10.2.2.256	172.16.10.25

Based on the subnet mask, which IP addresses could be grouped together into the same subnet?
What would their subnet ID be?
Is the address a legitimate host IP, a broadcast, or the subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/24	/28	/25	/29
Delta	1	16	128	
Octet	3 rd	4 th	4 th	
Example	0 – 255	0 – 15	0 – 127	
Ranges	0.0 – 0.255	16 – 31	128 – 255	
	1.0 – 1.255	32 – 47		
	2.0 – 2.255	128 - 143		
	3.0 – 3.255	160 - 175		
		...		
Subnet – even	192.168.2.0	192.168.1.0	10.2.2.127	172.16.10.0
1 st – odd	192.168.5.1	192.168.1.1	10.2.2.128	172.16.10.24
Last – even	192.168.2.255	192.168.1.13	10.2.2.190	172.16.10.31
B'cast - odd	192.168.2.28	192.168.1.14	10.2.2.192	172.16.10.30
	192.168.3.166	192.168.1.15	10.2.2.254	172.16.10.8
	192.168.2.20	192.168.1.16	10.2.2.255	172.16.10.26
	192.168.4.254	192.168.1.17	10.2.2.256	172.16.10.25

Based on the subnet mask, which IP addresses could be grouped together into the same subnet?
What would their subnet ID be?
Is the address a legitimate host IP, a broadcast, or the subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/24	/28	/25	/29
Delta	1	16	128	8
Octet	3 rd	4 th	4 th	4 th
Example	0 – 255	0 – 15	0 – 127	0 – 7
Ranges	0.0 – 0.255	16 – 31	128 – 255	8 – 15
	1.0 – 1.255	32 – 47		16 – 23
	2.0 – 2.255	128 - 143		24 – 31
	3.0 – 3.255	160 - 175		40 – 47
		...		80 - 87
				...
Subnet – even	192.168.2.0	192.168.1.0	10.2.2.128	172.16.10.0
1 st – odd	192.168.5.1	192.168.1.1	10.2.2.128	172.16.10.24
Last – even	192.168.2.255	192.168.1.13	10.2.2.190	172.16.10.31
B’cast - odd	192.168.2.28	192.168.1.14	10.2.2.192	172.16.10.30
	192.168.3.166	192.168.1.15	10.2.2.254	172.16.10.8
	192.168.2.20	192.168.1.16	10.2.2.255	172.16.10.26
	192.168.4.254	192.168.1.17	10.2.2.256	172.16.10.25

Task 4

- Based on the subnet mask, which IP addresses could be grouped together into the same subnet?
- What would their subnet ID be?
- Is the address a legitimate host IP, a broadcast, or the subnet ID?

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

/27

192.168.100.128
192.168.100.129
192.168.100.130
192.168.100.159
192.168.100.160
192.168.100.161
192.168.100.191

/26

192.168.100.0
192.168.100.65
192.168.100.130
192.168.100.254
192.168.100.200
192.168.100.191
192.168.100.192

/30

192.168.100.128
192.168.100.129
192.168.100.130
192.168.100.131
192.168.100.16
192.168.100.160
192.168.100.190

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/27	/26	/30
Delta	32		
Octet	4 th		
Example Ranges	0 – 31		
	32 – 63		
	64 – 95		
	128 – 159		
	...		
Subnet – even 1 st – odd Last – even B'cast - odd	192.168.100.128	192.168.100.0	192.168.100.128
	192.168.100.129	192.168.100.65	192.168.100.129
	192.168.100.130	192.168.100.130	192.168.100.130
	192.168.100.159	192.168.100.254	192.168.100.131
	192.168.100.160	192.168.100.200	192.168.100.16
	192.168.100.161	192.168.100.191	192.168.100.160
	192.168.100.191	192.168.100.192	192.168.100.190

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/27	/26	/30
Delta	32		
Octet	4 th		
Example Ranges	0 – 31 32 – 63 64 – 95 128 – 159 160 - 191		
Subnet – even 1 st – odd Last – even B'cast - odd	192.168.100.128	192.168.100.0	192.168.100.128
	192.168.100.129	192.168.100.65	192.168.100.129
	192.168.100.130	192.168.100.130	192.168.100.130
	192.168.100.159	192.168.100.254	192.168.100.131
	192.168.100.160	192.168.100.200	192.168.100.16
	192.168.100.161	192.168.100.191	192.168.100.160
	192.168.100.191	192.168.100.192	192.168.100.190

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/27	/26	/30
Delta	32	64	
Octet	4 th	4 th	
Example Ranges	0 – 31 32 – 63 64 – 95 96 – 223 ...	0 – 63 64 – 127 128 – 191 192 – 255	
Subnet – even 1 st – odd Last – even B'cast - odd	192.168.100.128 192.168.100.129 192.168.100.130 192.168.100.159 192.168.100.160 192.168.100.161 192.168.100.191	192.168.100.0 192.168.100.65 192.168.100.130 192.168.100.254 192.168.100.200 192.168.100.191 192.168.100.192	192.168.100.128 192.168.100.129 192.168.100.130 192.168.100.131 192.168.100.16 192.168.100.160 192.168.100.190

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/27	/26	/30
Delta	32	64	
Octet	4 th	4 th	
Example Ranges	0 – 31 32 – 63 64 – 95 96 – 223 ...	0 – 63 64 – 127 128 – 191 192 – 255	
Subnet – even 1 st – odd Last – even B'cast - odd	192.168.100.128 192.168.100.129 192.168.100.130 192.168.100.159 192.168.100.160 192.168.100.161 192.168.100.191	192.168.100.0 192.168.100.65 192.168.100.130 192.168.100.254 192.168.100.200 192.168.100.191 192.168.100.192	192.168.100.128 192.168.100.129 192.168.100.130 192.168.100.131 192.168.100.16 192.168.100.160 192.168.100.190

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/27	/26	/30
Delta	32	64	4
Octet	4 th	4 th	4 th
Example Ranges	0 – 31 32 – 63 64 – 95 96 – 223 ...	0 – 63 64 – 127 128 – 191 192 – 255	0 – 3 40 – 43 80 – 83 120 – 123 128 – 131
Subnet – even 1 st – odd Last – even B'cast – odd	192.168.100.128	192.168.100.0	192.168.100.128
	192.168.100.129	192.168.100.65	192.168.100.129
	192.168.100.130	192.168.100.130	192.168.100.130
	192.168.100.159	192.168.100.254	192.168.100.131
	192.168.100.160	192.168.100.200	192.168.100.16
	192.168.100.161	192.168.100.191	192.168.100.160
	192.168.100.191	192.168.100.192	192.168.100.190

Task 5

- Based on the subnet mask, which IP addresses could be grouped together into the same subnet?
- What would their subnet ID be?
- Is the address a legitimate host IP, a broadcast, or the subnet ID?

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

/20

192.168. 0.1
192.168. 32.1
192.168. 160.1
192.168. 174.1
192.168. 175.254
192.168. 176.1
192.168. 192.1

/22

192.168. 4.0
192.168. 8.1
192.168. 12.1
192.168. 44.0
192.168. 44.1
192.168. 44.255
192.168. 46.3

/21

192.168. 8.1
192.168. 32.1
192.168. 40.1
192.168. 80.1
192.168. 81.1
192.168. 87.1
192.168. 88.1

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/20	/22	/21
Delta	16		
Octet	3 rd		
Example Ranges	0.0 – 15.255 32.0 – 63.255 160.0 – 175.255 176.0 – 191.255 ...		
Subnet – even 1 st – odd Last – even B'cast - odd	192.168. 0.1	192.168. 4.0	192.168. 8.1
	192.168. 32.1	192.168. 8.1	192.168. 32.1
	192.168. 160.1	192.168. 12.1	192.168. 40.1
	192.168. 174.1	192.168. 44.0	192.168. 80.1
	192.168. 175.254	192.168. 44.1	192.168. 81.1
	192.168. 176.1	192.168. 44.255	192.168. 87.1
	192.168. 192.1	192.168. 46.3	192.168. 88.1

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/20	/22	/21
Delta	16	4	
Octet	3 rd	3 rd	
Example Ranges	0.0 – 15.255 32.0 – 63.255 160.0 – 175.255 176.0 – 191.255 ...	0.0 – 3.255 4.0 – 7.255 40.0 – 43.255 44.0 – 47.255 160.0 – 163.255	
Subnet – even 1 st – odd Last – even B'cast - odd	192.168. 0.1	192.168. 4.0	192.168. 8.1
	192.168. 32.1	192.168. 8.1	192.168. 32.1
	192.168. 160.1	192.168. 12.1	192.168. 40.1
	192.168. 174.1	192.168. 44.0	192.168. 80.1
	192.168. 175.254	192.168. 44.1	192.168. 81.1
	192.168. 176.1	192.168. 44.255	192.168. 87.1
	192.168. 192.1	192.168. 46.3	192.168. 88.1

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

128 64 32 16 8 4 2 1
7 6 5 4 3 2 1 0

	/20	/22	/21
Delta	16	4	8
Octet	3 rd	3 rd	3 rd
Example Ranges	0.0 – 15.255 32.0 – 63.255 160.0 – 175.255 176.0 – 191.255 ...	0.0 – 3.255 4.0 – 7.255 40.0 – 43.255 44.0 – 47.255 160.0 – 163.255	0.0 – 7.255 40.0 – 47.255 80.0 – 87.255 120.0 – 127.255 ...
Subnet – even 1 st – odd Last – even B'cast - odd	192.168. 0.1	192.168. 4.0	192.168. 8.1
	192.168. 32.1	192.168. 8.1	192.168. 32.1
	192.168. 160.1	192.168. 12.1	192.168. 40.1
	192.168. 174.1	192.168. 44.0	192.168. 80.1
	192.168. 175.254	192.168. 44.1	192.168. 81.1
	192.168. 176.1	192.168. 44.255	192.168. 87.1
	192.168. 192.1	192.168. 46.3	192.168. 88.1

Task 6

- Based on the subnet mask, which IP addresses could be grouped together into the same subnet?
- What would their subnet ID be?
- Is the address a legitimate host IP, a broadcast, or the subnet ID?

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

/18	/23	/17
192.168.5.1	192.168.100.0	172.30.0.0
192.168.126.129	192.168.101.65	172.30.1.0
192.168.130.130	192.168.101.254	172.30.126.0
192.168.192.159	192.168.199.200	172.30.127.254
192.168.192.0	192.168.200.191	172.30.128.1
192.168.200.161	192.168.201.20	172.31.128.2
192.168.254.255	192.168.202.40	172.30.99.255

Which IP addresses could be grouped together into the same subnet based on the subnet mask?
 What would their subnet ID be?
 Is the address a legitimate host IP, a broadcast, or a subnet ID?

128 64 32 16 8 4 2 1
 7 6 5 4 3 2 1 0

	/18	/23	/17
Delta	64		
Octet	3 rd		
Example Ranges	0.0 – 63.255 64.0 – 127.255 128.0 – 191.255 192.0 – 255.255		
	192.168.5.1	192.168.100.0	172.30.0.0
	192.168.126.129	192.168.101.65	172.30.1.0
	192.168.130.130	192.168.101.254	172.30.126.0
	192.168.192.159	192.168.199.200	172.30.127.254
	192.168.192.0	192.168.200.191	172.30.128.1
	192.168.200.161	192.168.201.20	172.31.128.2
	192.168.254.255	192.168.202.40	172.30.99.255

Subnet – even
 1st – odd
 Last – even
 B'cast - odd

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/18	/23	/17
Delta	64	2	
Octet	3 rd	3 rd	
Example Ranges	0.0 – 63.255 64.0 – 127.255 128.0 – 191.255 192.0 – 255.255	0.0 – 1.255 2.0 – 3.255 10.0 – 11.255 16.0 – 17.255 ...	
	192.168.5.1 192.168.126.129 192.168.130.130 192.168.192.159 192.168.192.0 192.168.200.161 192.168.254.255	192.168.100.0 192.168.101.65 192.168.101.254 192.168.199.200 192.168.200.191 192.168.201.20 192.168.202.40	172.30.0.0 172.30.1.0 172.30.126.0 172.30.127.254 172.30.128.1 172.31.128.2 172.30.99.255

Subnet – even
1st – odd
Last – even
B'cast - odd

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/18	/23	/17
Delta	64	2	
Octet	3 rd	3 rd	
Example Ranges	0.0 – 63.255 64.0 – 127.255 128.0 – 191.255 192.0 – 255.255	0.0 – 1.255 2.0 – 3.255 10.0 – 11.255 16.0 – 17.255 ...	
	192.168.5.1 192.168.126.129 192.168.130.130 192.168.192.159 192.168.192.0 192.168.200.161 192.168.254.255	192.168.100.0 192.168.101.65 192.168.101.254 192.168.199.200 192.168.200.191 192.168.201.20 192.168.202.40	172.30.0.0 172.30.1.0 172.30.126.0 172.30.127.254 172.30.128.1 172.31.128.2 172.30.99.255

Subnet – even
1st – odd
Last – even
B'cast - odd

Which IP addresses could be grouped together into the same subnet based on the subnet mask?

What would their subnet ID be?

Is the address a legitimate host IP, a broadcast, or a subnet ID?

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

	/18	/23	/17
Delta	64	2	128
Octet	3 rd	3 rd	3 rd
Example Ranges	0.0 – 63.255 64.0 – 127.255 128.0 – 191.255 192.0 – 255.255	0.0 – 1.255 2.0 – 3.255 10.0 – 11.255 16.0 – 17.255 ...	0 – 127 128 - 255
Subnet – even 1 st – odd Last – even B'cast - odd	192.168.5.1	192.168.100.0	172.30.0.0
	192.168.126.129	192.168.101.65	172.30.1.0
	192.168.130.130	192.168.101.254	172.30.126.0
	192.168.192.159	192.168.199.200	172.30.127.254
	192.168.192.0	192.168.200.191	172.30.128.1
	192.168.200.161	192.168.201.20	172.31.256.2
	192.168.254.255	192.168.202.40	172.30.99.255

